

C. Counting Relative Lists

time limit per test: 1 second

memory limit per test: 256 megabytes

You are given a task to count how many lists of N positive integer numbers, each less than or equal to M , can be formed such that adjacent numbers in the list are relatively prime. Two numbers are relatively prime if their greatest common divisor (GCD) is 1. This problem combines number theory with combinatorial counting, challenging you to explore the intricacies of prime relationships between numbers within a constrained set.

Input

Input consists of a single line containing two integers N and M ($1 \leq N \leq 10^8$) ($1 \leq M \leq 100$), representing the length of the lists and the maximum possible value of each element in the lists, respectively.

Output

Output a single integer, the total number of lists that can be formed under the given conditions, because the number can be very large print it modulo $10^9 + 7$.

Examples

input	Copy
2 3	
output	Copy
7	

input	Copy
2 10	
output	Copy
63	

First Masters Championship
LATAM 2024

Finished

Practice



→ **About Contest**



Primer Campeonato de Masters de Latinoamérica 2024

[Contest website](#)

→ **Virtual participation** 

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You can clone this contest to a mashup.

Clone Contest

→ **Submit?**

Language: GNU G++23 14.2 (64 bit, ms) 

Choose file:

Choose File No file chosen

Submit

→ **Last submissions**

Submission	Time	Verdict
336697402	Sep/03/2025 04:02	Happy New Year!